

Apply to be a Science Corps Fellow - Entry #3343

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Indian
When did you or do you expect to complete your PhD?
2025
Please tell us briefly about your university education (institution, city, country, year)
I'm pursuing a PhD in experimental nuclear and particle physics at Case Western Reserve University (CWRU) in Cleveland, Ohio, USA. My research focuses on finding alternatives to xenon for use in large, underground gas detectors that search for rare signals like neutrinoless double beta decay—a key to understanding why the universe is made mostly of matter, not antimatter. My day-to-day work involves instrumentation and simulating interactions between radiation and matter to improve future detectors. I transferred to CWRU from SUNY Buffalo in Fall 2020 and have been actively involved in research here. Before coming to the US, I earned my undergraduate degree in Physics from the University of Delhi. I completed my Master's in Physics at the Indian Institute of Technology in Guwahati, India, where I worked with optical tweezers and simulated image reconstruction using masked diffraction patterns.
Why would you like to be a Science Corps Fellow?
Over the past few years, I've realized that while pushing the boundaries of science is thrilling, what matters just as much is how that science ripples outwards. As a PhD student working on experimental particle and nuclear physics, I've spent countless hours refining detectors, running simulations, and trying to play my part in understanding the universe at its most fundamental level. But some of the most meaningful moments in my journey have come from outside the lab—when I've been able to help someone else see themselves in science. During my PhD, I took a high school student under my wing—someone curious about physics but unsure about what it meant beyond textbooks and classroom lectures. I regularly respond to messages from students on LinkedIn, especially those navigating fellowship and scholarship

applications. I provide honest advice, share resources, and try to be the person I wish I had been when I started. These conversations are often short, but they leave a lasting impression—on both sides. It has shown me that impact doesn't have to be loud to be meaningful.

That's why the Science Corps Fellowship speaks to me—it blends science, service, and global impact personally and meaningfully. I want to take the technical skills I've developed—understanding radiation, working with detectors, solving challenging problems—and use them in a setting where the focus isn't discovery but empowerment. I want to help students in underserved communities see that science isn't reserved for the chosen few, that they are competent, and that they don't need to wait for someone else to hand them the tools—they can build their own.

Growing up and studying in India, I know first-hand how access to quality science education can be limited, especially outside major cities. I was lucky to have mentors who encouraged me to reach higher, and I carry that within me. During my time in the US, I've often thought about how to pay that forward—not just by doing good science but by sharing it meaningfully with others. Teaching internationally through Science Corps would be a way for me to give back and help build local capacity in places where talent and curiosity are abundant, but resources might not be.

As I look ahead to graduating in 2025, I am considering aligning my career with values beyond prestige or technical achievement. I'm particularly drawn to the idea that this Fellowship creates an impact that outlives the time we spend teaching because of its short-term but high-impact nature. While exploring long-term opportunities in national labs or industry, I crave a moment of connection before jumping into the next big thing. I want to spend this transition period doing something grounded and human, recharge, and get in touch with my 'why' for choosing this career. By developing a curriculum and supporting local teachers, I hope to leave behind something sustainable—resources and confidence that continue to grow after I leave.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Throughout my academic journey, I've made a conscious effort to extend the impact of my work beyond the boundaries of higher education. While physics research is the core of my professional life, I've always been driven by a side quest involving reaching out to people who may never set foot in a university classroom but deserve access to science, curiosity, and mentorship just as much as anyone else.

While pursuing a Masters in Physics at the Indian Institute of Technology Guwahati, India, I spent a few weekends volunteering as a teacher for school children (ages 5-15) of campus staff and families in the nearby village. This informal, student-led initiative sought to bridge the educational gap for kids with little to no support outside the minimum provided in local schools. We taught math, science, and basic English literacy. Still, more than that, we aimed to build a space where learning felt exciting and personal—where these students could see themselves as capable, intelligent, and full of potential. This experience shaped my belief that science outreach must be relational, rooted in community, and ongoing.

That early work has stayed with me and continues to guide how I view my responsibilities as a scientist. As I move through graduate school, I remain committed to mentoring and service that connects with people at the grassroots

level. I actively mentor early-career students in physics, especially those navigating the academic world without the benefit of legacy or insider knowledge. I am working with a high school student in Cleveland who wanted a taste of college life and had parents who never went to college. I want to serve as a career counselor and mentor for those curious about science who may not yet have a clear path to pursue it. I believe meaningful change often begins with a conversation, and I want to be available for those conversations—with students, families, and communities.

My research has taken me to conferences and academic forums. Still, I've come to believe that the most transformative stories we tell about science don't happen in lecture halls or poster sessions—they unfold in places where curiosity is still raw and unfiltered. I remember the first time I helped set up a small telescope with friends and watched a star streak across the lens in the time it took us to align the room. It blew my mind. I had known for years that the Earth moves, but feeling that motion in real time was entirely different. Last month, I focussed the university telescope on Jupiter and saw three of its moons. Even at 28, it made me feel like a child in a candy store. I wonder how it would have changed my life to see that as a 15-year-old, and it's what motivates me to reach out to students who might not have access to such moments otherwise. I want to be part of building a space where kids and young adults get to feel that wonder, too, and maybe see a future for themselves in science.

Where do you imagine yourself professionally in the future? Check all that apply.

Doing science in industry

Administrative role in industry or academia

In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

In the next few years, I see myself working as an experimental physicist in a national lab or a medical device industry. I want to keep building detectors and working on projects, pushing the boundaries of knowledge. I recently became very interested in imaging and medical research. It is a field that I was exposed to only after coming to the United States, and I have discovered new pathways to apply my training and skills in radiation physics to the medical field, where experts use radiation to diagnose and treat cancers.

At the same time, I enjoy organizing and helping people work together. I'm open to roles beyond research—like managing a project, coordinating between teams, or helping run a division in industry or academia. I like understanding how systems work, solving problems that aren't just technical, and assisting different groups to communicate. We achieve more when we work together. In today's time, when science is no longer an individualistic endeavor, we need the power of the people to come together synergistically. I want to be respected enough in my field to be asked to head an organization. I aspire to be skilled enough to do the role justice and be a team leader.

In the long run—say, ten years from now—I see myself in a role that combines science, leadership, and public outreach. I want to help decide where science is headed and make sure more people get to be part of it. Whether that means leading an outreach program, managing a research group, or even starting my nonprofit, I want to build something that makes science more open and welcoming. I'm especially interested in working with students who don't come from traditional academic backgrounds and might not think science is for them.

These moments—teaching a child how to use a telescope and explaining why the stars move—are just as crucial as any paper or conference. They remind me why I do science in the first place and who it's for. My career might take many shapes, but no matter where I end up, I want to keep reaching out and helping others see that science is something they can be a part of too.

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An email list in your institution

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